

Sociology N100 Syllabus, Summer 2026

Housing Precarity and Displacement: Racial and Gender Inequality in Gentrification and Eviction

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Housing precarity is how vulnerable a household is to losing stable housing when hit by a sudden change, such as job loss or rising housing costs (Pendall et al., 2012). The more precarious a household, the greater its risk of residential displacement: the involuntary relocation of residents from their homes or neighborhoods due to factors beyond their control. This can occur when households are forced to move because of evictions, rent increases, building demolitions, or other changes that make their housing unaffordable or uninhabitable. The consequences of displacement are dire, leading to long-term housing and financial instability; worse health outcomes; and reinforces racial, gender, and economic disparities. Today, a large share of the United States is facing housing precarity and are at great risk of displacement. Roughly 50% of renters are rent-burdened (paying more than 30% of their income to rent), about 1/3rd (32%) of adults are likely to miss next month's rent or mortgage payment, and homelessness is higher now than it was during the 2008 Great Recession. Institutions are determined to understand the prevalence and risk of displacement because it affects both households at the micro level and shapes cities and regions at the macro level. But how do we measure displacement and what can be done about it?

This course takes up those questions through a sociological lens and data-driven analysis. Using the U.S. Census and R programming, you will learn to measure displacement and its drivers: quantify racial inequality, track segregation patterns, and map gentrification's impact on vulnerable communities. You will build practical computational social science skills, from data wrangling and spatial analysis to statistical modeling and visualization. A central goal is to use data as public scholarship: to show a policymaker or community stakeholder how to understand a common trend or problem and help inform legislation, using sociology and theory to turn evidence into a clear, honest case you can make to real people.

Through weekly lectures, coding labs, and interactive online modules, each week builds toward a final group project. You will complete two short assignments that ladder into that project, and by term's end you will create dynamic maps and interactive dashboards that reveal the stories behind eviction trends, rent burden disparities, and neighborhood change. Designed for beginners in coding, this fully online course emphasizes real-world applications, with case studies ranging from urban displacement in coastal cities to rural housing instability.

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Sociology N100, Section 002 (Class #13195) | 2 units | Summer Session D 2026

Tuesdays & Thursdays, 5-7pm | July 7 to August 13, 2026 | Online via Zoom

Links: [Course Site](#) | [Class Zoom](#) | [bCourses](#) | [Discussion](#)

Office Hours: schedule at cal.com/timthomas; evenings and weekends also available by email request.

Course Schedule

Week	Dates	Topic
1	Jul 7 & 9	Intro to housing precarity & displacement; R/RStudio setup; Census/ACS basics; how to use AI
2	Jul 14 & 16	Quantifying racial inequality; tidycensus wrangling (Read: Zuk et al. 2015; Walker Pref-Ch2)
3	Jul 21 & 23	Segregation measures and mapping with sf/tigris; Assignment 1 due Mon Jul 27 (Read: Walker Ch3-4)
4	Jul 28 & 30	Gentrification and neighborhood change; rent burden (Read: Walker Ch5-6)
5	Aug 4 & 6	Eviction trends + ggplot; interactive maps; Assignment 2 due Mon Aug 3 (Read: Bates 2013)
6	Aug 11 & 13	Dashboards (Quarto) and final project presentations (last class Aug 13)

Final project due Friday, August 14, 2026.

Course Culture

This course brings together students from diverse backgrounds to examine housing inequality through data analysis in R. We create a collaborative, listening-centered environment where all voices are valued and where we approach housing data with critical consciousness about systemic racism and gentrification's human impacts.

Our shared commitments:

- **Listen first** - We begin with the experiences of communities affected by housing displacement before analyzing statistics
- **Intersectional analysis** - We examine how race, gender, and class intersect to create different experiences of housing precarity
- **Respectful dialogue** - We engage difficult conversations about inequality with professionalism and care for one another
- **Collective learning** - We develop data skills together, supporting each other's technical and critical learning

Creating brave space: We recognize that housing data reveals uncomfortable truths about displacement and discrimination. We commit to supporting classmates who may be personally affected by housing insecurity while holding ourselves accountable for learning that serves rather than extracts from communities.

If you have concerns about classroom environment or need support, please reach out to me to discuss how we can better foster inclusive learning for everyone.

Materials

The primary materials for the course are available on this website. I'll teach you everything you need to know!

RStudio

The software that we'll be using for our data analysis is the free and open-source language called R that we'll be interacting with via software called RStudio. As a Berkeley student, you have your own version of RStudio waiting for you on a special [data hub](#). This data hub syncs the class materials, giving you access to most of what you need. Most students taking this course have no experience programming; I'll teach you everything you need to know!

Workload

- 4 hours of class time (2 hours of lecture, 2 hours of lab)
- expect 12 hours of work outside of class after week 1

Instruction Mode

This course will be held on Zoom with recordings posted after each class.

Course Requirements and Assignments

Your grade has three components, which have the following weights:

1. Class participation 20%
2. Assignments (#1 = 20%, #2 = 20%) 40%
3. Final group project 40%

Class participation (20%)

Participation in class is an important part of your learning experience and a vital skill in the working world. Participation in class also gives me richer information about your grasp of the class material and helps me make adjustments where necessary. Attendance makes up 10% for class participation and the remaining 10% is based on your participation - in-class quizzes, contributing to class discussions, asking questions, offering up potential solutions to problems, presenting a solution to an assignment, volunteering to present, and overall engagement and enrichment of the class.

A good contribution to class discussion is a comment that possesses one or more of the following properties. (1) It offers a different and unique, but relevant, insight to the issue; moves the discussion and analysis forward to generate new insights. (2) It builds on the preceding discussion; relates to a personal anecdote or experience in a way that helps to illuminate the ideas being discussed. (3) It uses logic, evidence, and creative thinking (argument), so that it is more than merely an expression of an opinion or feeling (assertion).

Assignments (40%).

For each assignment you will submit:

- R code script that is replicable by the grader,
- a brief write up explaining your assignment with plots and context, and
 - Include one or more introductory paragraphs summarizing what you are analyzing and your overall findings.
 - Text and plots walking the reader through a more detailed understanding of what you found. Make sure to describe your plots.
 - Summary conclusion and analytic insights (i.e., what the data is telling you).
- URL links to any and all AI usage.
 - In your code, put the URL in comments next to the relevant area.
 - In your write up, put the URL in footnotes near the section or text in which you used AI.
 - *See AI policy below for more info.*

These assignments will be graded on using the scale below. Anything above a D will pass. No late assignments will be accepted. You can volunteer to present a 5-minute demo of your assignment in class.

Final group project (40%).

For your final group project you and 2-3 of your peers will choose (or be assigned) a city or county and come up with a research question you can answer with one of the datasets from the class that tests at least two hypotheses about the disparate impact or unique problem of displacement in that area. You will need to do some research about your chosen area and may even need to do some exploratory analysis of the area to see what might be a suitable research question and driver of displacement. For example, in Marin County, CA high rents are displacing the county's workforce-age adults and young families, resulting in school closures and deterring needed residents to keep the county sustainable. Another example is Salt Lake City's most affordable neighborhoods are also experiencing high rates of displacement, meaning there is no where for displaced residents to move to that is affordable. Find something that would be compelling for the city or county to know about the problem of displacement in their area. Write for a policymaker or community stakeholder: your job is to help them understand the trend or problem and act on it. **Bonus points** for a strong, well-supported set of policy recommendations for the problem at hand.

- You will submit a final report, R code, and a short slide presentation (no longer than 10 slides) showcasing your question, your methods and key results.
- On the last day you will present your project to the class.

Course Grading

A+ 97+ | A 93-96 | A- 90-92

B+ 87-89 | B 83-86 | B- 80-82

C+ 77-79 | C 73-76 | C- 70-72

D+ 67-69 | D 63-66 | D- 60-62

F 0-59

Late work policy

No late work will be accepted except for documented DSP accommodations or arrangements made with the instructor in advance.

AI Policy

AI is allowed and encouraged in this course as it is crucial to learn how to use these tools effectively for your future work. However, it is vital to recognize both the strengths and limitations of AI to avoid jeopardizing your projects and future career. Here are some key considerations:

Understanding AI's Role

1. *Information Synthesis:* AI excels at synthesizing information but does not create new knowledge. It serves as a broad assistant for research, writing improvement, and coding tasks.
2. *Limitations of Training Data:* AI models are constrained by their training data and cannot generate truly novel ideas or content beyond this scope.

3. *Potential for Errors*: AI hallucinations and misinterpretations are real concerns. Always critically evaluate and verify AI-generated content.

Effective AI Usage

1. *Prompt Engineering*: The quality of AI output heavily depends on the quality of input. Craft clear, specific prompts to get the best results.
2. *Human Expertise*: You are the domain expert. Ensure that AI is working for you and aligning with your goals and expectations.
3. *Fact-Checking*: Always verify AI-generated information, especially for recent events or specialized topics.

Ethical Considerations

1. *Attribution*: When using AI-generated content, properly attribute it in your work.
2. *Academic Integrity*: Use AI as a tool for learning and productivity, not as a substitute for your own critical thinking and analysis.
3. *Bias Awareness*: Be mindful of potential biases in AI-generated content and strive for diverse perspectives.

Be vigilant about model changes moving forward: As AI advances, be aware of both improvements to models as well as possible problems (i.e., increasing censorship, how it was trained, developer intent, etc.).

Which AI tool you use, and the one hard rule

You may use any AI assistant you like in this class **as long as it can produce a public, shareable link to your conversation that I can open and read without logging in**. You will submit those links with your work (URLs in your code comments and footnotes in your writeup). I read them to understand how you worked and to help you get better at it. If a tool cannot give me a link I can open, you cannot use it for graded work.

Any of these meet the bar, and a **free account is enough** (you never need a paid plan for this course). If you are not sure where to start, use Perplexity:

- **Perplexity** (recommended) searches the web, shows its sources, and lets you compare several models. Share a Thread by setting it to “Anyone with the link.”
- **ChatGPT**: use the **Share** button to make a public link.
- **Google Gemini**: use **Share** to make a public link.
- **Claude**: use **Share** to make a public link.

Two things that trip people up:

1. **Sign in with a personal account, not your CalNet/Berkeley login**. Berkeley’s free versions of Gemini and Copilot run in a protected mode that turns public sharing *off*. We only use public Census data in this class, so a personal account is the right choice here.
2. **Test your link before you submit it**. Paste it into a private/incognito browser window. If it shows your whole conversation without asking you to log in, you are good. If it does not, I cannot grade it.

Free AI access through Berkeley

Berkeley gives every student free access to some AI tools through the campus AI hub, currently the **Google Gemini app** and **Microsoft Copilot**, signed in with your CalNet (Berkeley) account. See the campus list, and the current set of tools, at <https://ai.berkeley.edu/tools-services/licensed-ai-tools>. These are a good choice for general use and are the safer option for any sensitive or private information.

One catch for *this* class: those Berkeley accounts run in a privacy-protected mode that usually turns public link-sharing **off**. Since your graded work here requires a link I can open, use a **personal** account (listed above) when you need to share a conversation. We only work with public Census data, so a personal account is fine for our purposes.

(These offers and free tiers change often. Re-check the tool’s “share” help page if a link will not generate.)

Responsible AI Use

1. *Transparency*: Be open about your use of AI tools in your work.
2. *Iterative Approach*: Use AI as part of an iterative process. Start with creating some content and then ask AI to refine and improve results with your own expertise.
3. *Skill Development*: Focus on developing skills in prompt engineering, critical evaluation, and effective integration of AI-generated content.

Reporting Issues

If you encounter any problems or have concerns about AI use in the course, please report them to the instructor promptly.

By following these guidelines, you’ll be better equipped to harness the power of AI while maintaining the integrity and quality of your academic work in computational social science.

Code of Conduct

Students and the instructor will abide by the [UC Berkeley Code of Conduct](#).

Primary Datasets & Tools

The American Community Survey (ACS) is an ongoing survey conducted by the U.S. Census Bureau that provides vital information about the United States and its people. Released annually, the ACS covers a broad range of topics including social, economic, housing, and demographic characteristics of the U.S. population. Some of the primary variables used to measure displacement that are derived from the ACS include housing characteristics, rent, racial and ethnic compositions, incomes, and migration.

tidycensus is an R package that provides a user-friendly interface to access ACS data directly within the R environment. It allows users to retrieve ACS data in a tidy format, ready for analysis with tidyverse tools. With tidycensus, you can easily download ACS data for various geographic levels (e.g., states, counties, census tracts) and years, and include spatial data for mapping purposes.

Disabled Student Program (DSP)

UC Berkeley is committed to creating a learning environment that meets the needs of its diverse student body including students with disabilities. If you anticipate or experience any barriers to learning in this course, please feel welcome to discuss your concerns with me.

If you have a disability, or think you may have a disability, you can work with the Disabled Students' Program (DSP) to request an official accommodation. The Disabled Students' Program (DSP) is the campus office responsible for authorizing disability-related academic accommodations, in cooperation with the students themselves and their instructors. You can find more information about DSP, including contact information and the application process here: dsp.berkeley.edu. If you have already been approved for accommodations through DSP, please meet with me so we can develop an implementation plan together."

Students who need academic accommodations or have questions about their accommodations should contact DSP, located at 260 César Chávez Student Center. Students may call 642-0518 (voice), 642-6376 (TTY), or e-mail dsp@berkeley.edu(link sends e-mail).

Below are some campus resources that may be helpful for you:

- [Berkeley Supportal- great place to start](#)
- [Disabled Students' Program](#)
- [University Health Services](#)
- [UCB Path to Care](#)
- [Student Learning Center](#)
- [Berkeley International Office](#)
- [Ombuds Office for Students and Postdoctoral Appointees](#)
- [Gender Equity Resource Center](#)
- [Center for Educational Justice & Community Engagement](#)
- [UHS Counseling and Psychological Services \(CAPS\)](#)
- [Campus Academic Accommodations Hub](#)
- [ASUC Student Advocate's Office](#)
- [Basic Needs Center](#)
- [ASUC Mental Health Resources Guide](#)